

LIGHT PORTALS

HERM & AARON GREIDER

Proposal for the 2026 Cincinnati Nature Center
Light in the Forest event

Contact

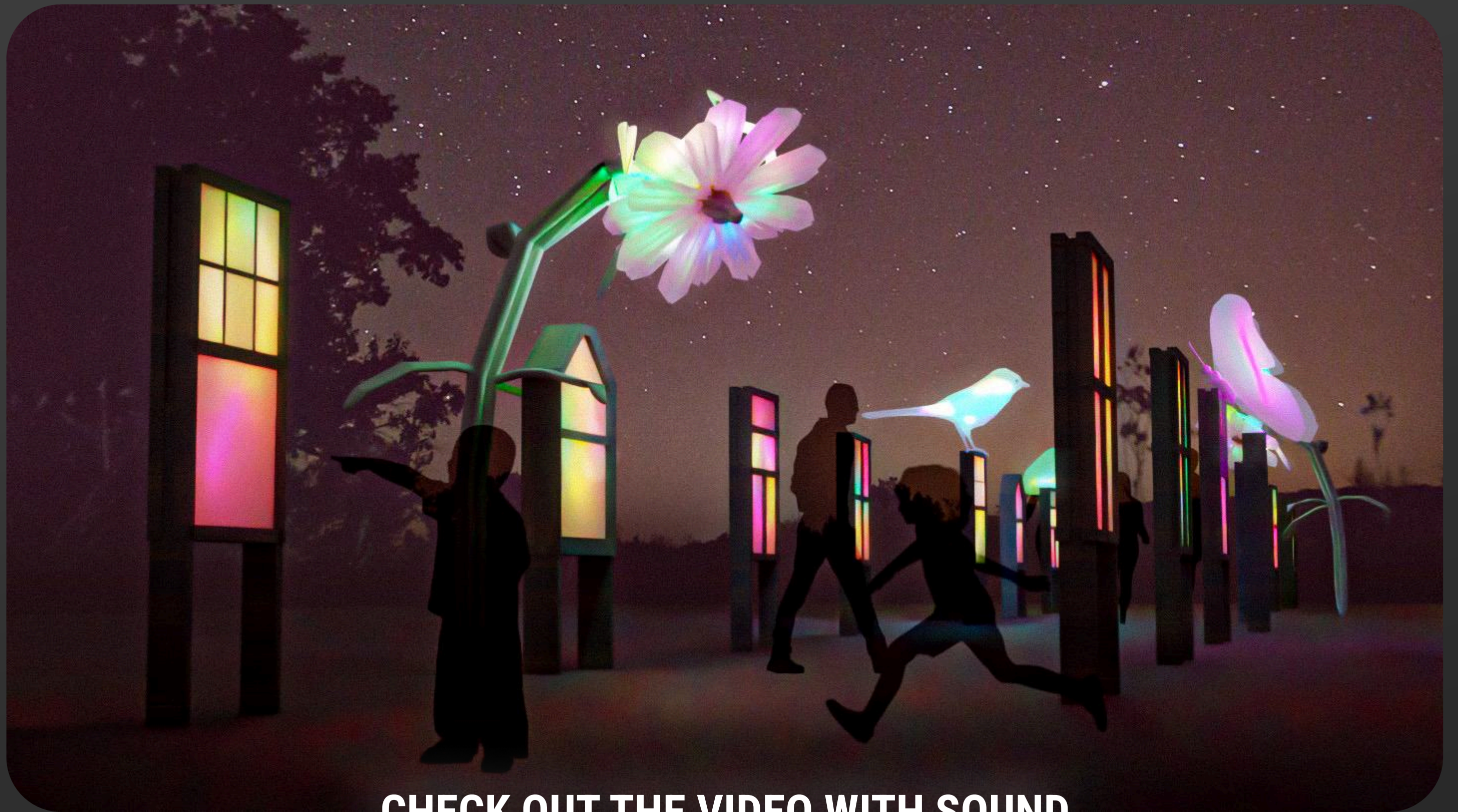
937-271-5532 call or text
herm.greider@gmail.com
4403 Erie Ave #3
45227 Cincinnati OH

2026 Proposal

light.greider.org/cnc2026

2025 Recap

light.greider.org



CHECK OUT THE VIDEO WITH SOUND

THE LIGHT PORTALS ARE BACK!

As you walk the nature path, soft chimes and flickering forest specters respond to your movements, creating a surreal, immersive experience. This year, the installation magnifies nature through the addition of larger-than-life weatherproof papier-mâché sculptures. These animal figures emerge from the shadows, their handcrafted textures catching the light and distorting the sense of scale.

By blending these massive organic forms with the glow of the LEDs, the Portals feel malleable and alive—inviting the audience to see the natural world not just as a backdrop, but as a towering, vibrant presence.

ABOUT THE PORTALS

The installation will be a construction of wooden frames with programmable multicolor LED strips sealed between two panes and fully waterproofed. Each LED responds to data from proximity sensors, changing hue, intensity, and pattern to bring the illuminated display to life. A programmable synthesizer will adjust notes and timbres based on visitor movement, fading tracks in and out and shaping waveforms for a more immersive experience. Each portal will feature a single speaker on the top, creating a compact surround sound setup.

IMPROVEMENTS

The first year was a little rough around the edges and we have several improvements prepared to polish our project up.

Larger than Life Nature Motifs

This year, keeping in the spirit of the portals, we will feature large internally illuminated paper mâché flora and fauna lanterns. We have proof of concept for weatherproofing, structure, and light transmission allowing each object to function much like each Portal does.

More Portals!

The Light Portals are a modular design. Since we'll be building off of our previous year's work, we can outfit the old Portals with new electronics and add up to 8 additional portals for a total of 16.

Let's chat about it!

Synchronized Light and Sound

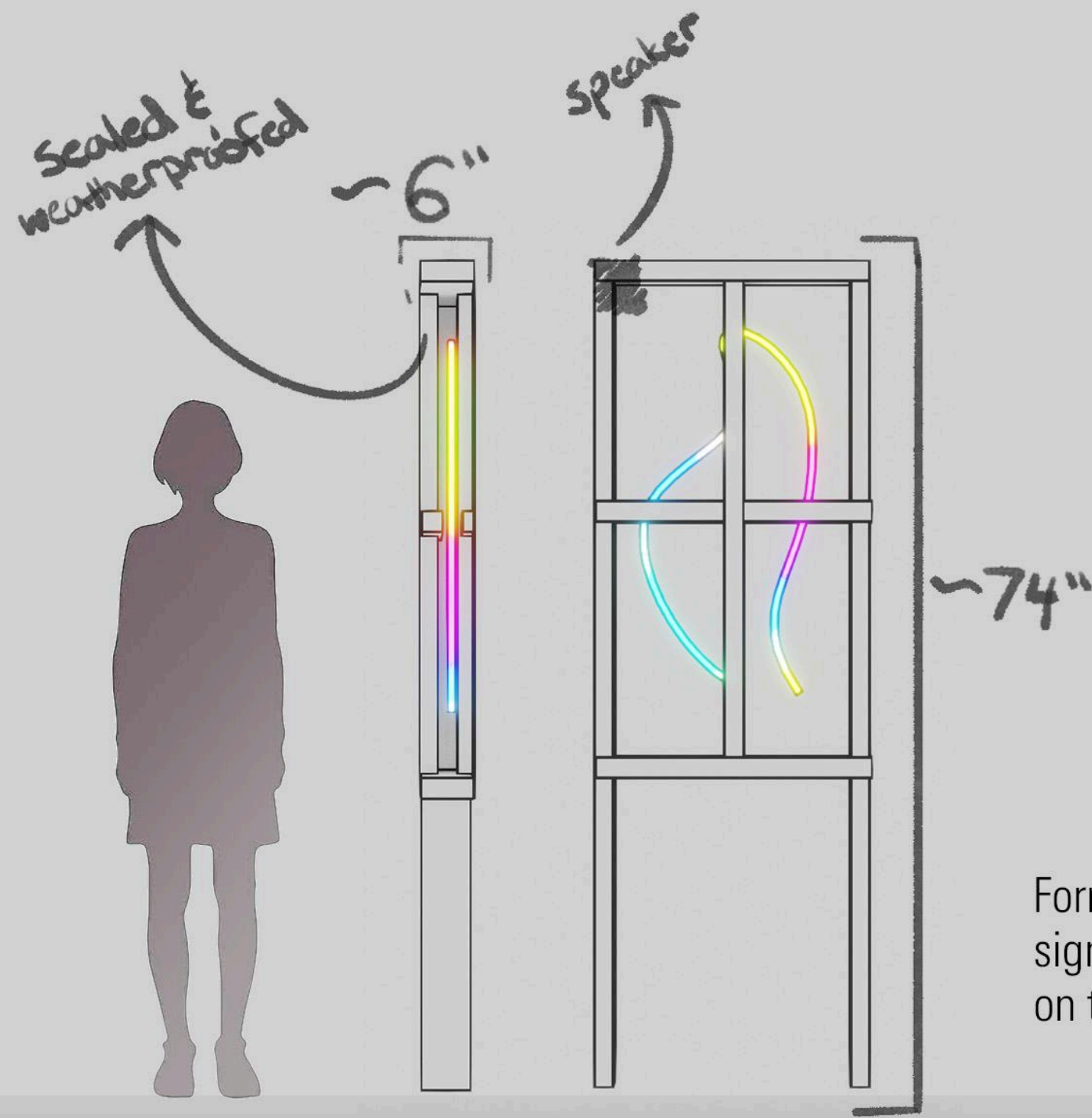
Sync was tougher than we thought! We've taken our down time to iterate and we have it prepped and ready. This year our music and lights will be synchronized across Portals, and will have new features such as motion and speed tracking for continuous and responsive lighting effects.

Lightweight Design

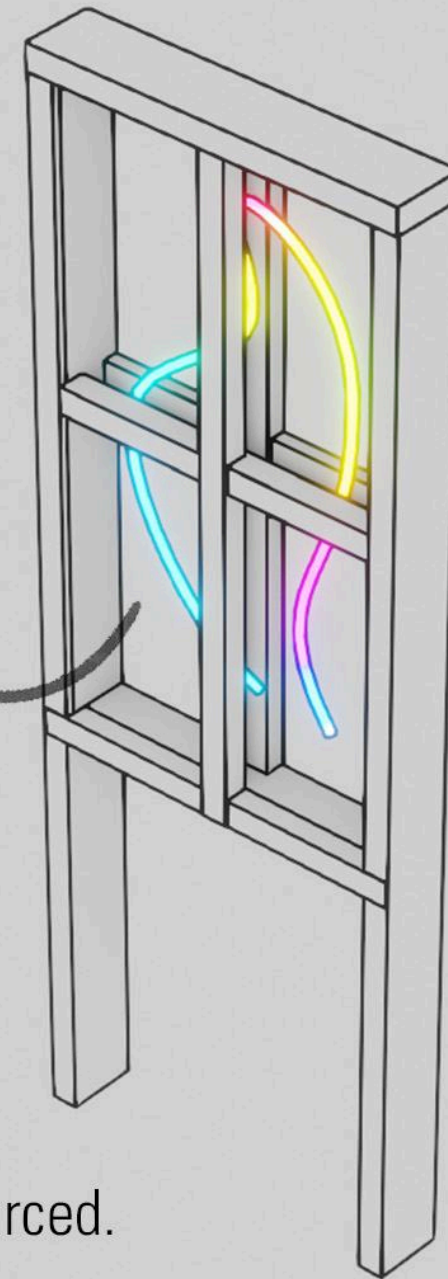
We'll build any additional Portals with smaller wooden frames and plexiglass to make each Portal lighter and more mobile.

Reliable Electronics

We assembled our first Portals with each one a remarkable snowflake. For 2026, we'll normalize and simplify the electronics to permit the addition of more portals with the same rock solid reliability as last year, but with less time-consuming calibration on each portal. We'll also upgrade our power system to avoid the spider web of extension cords.

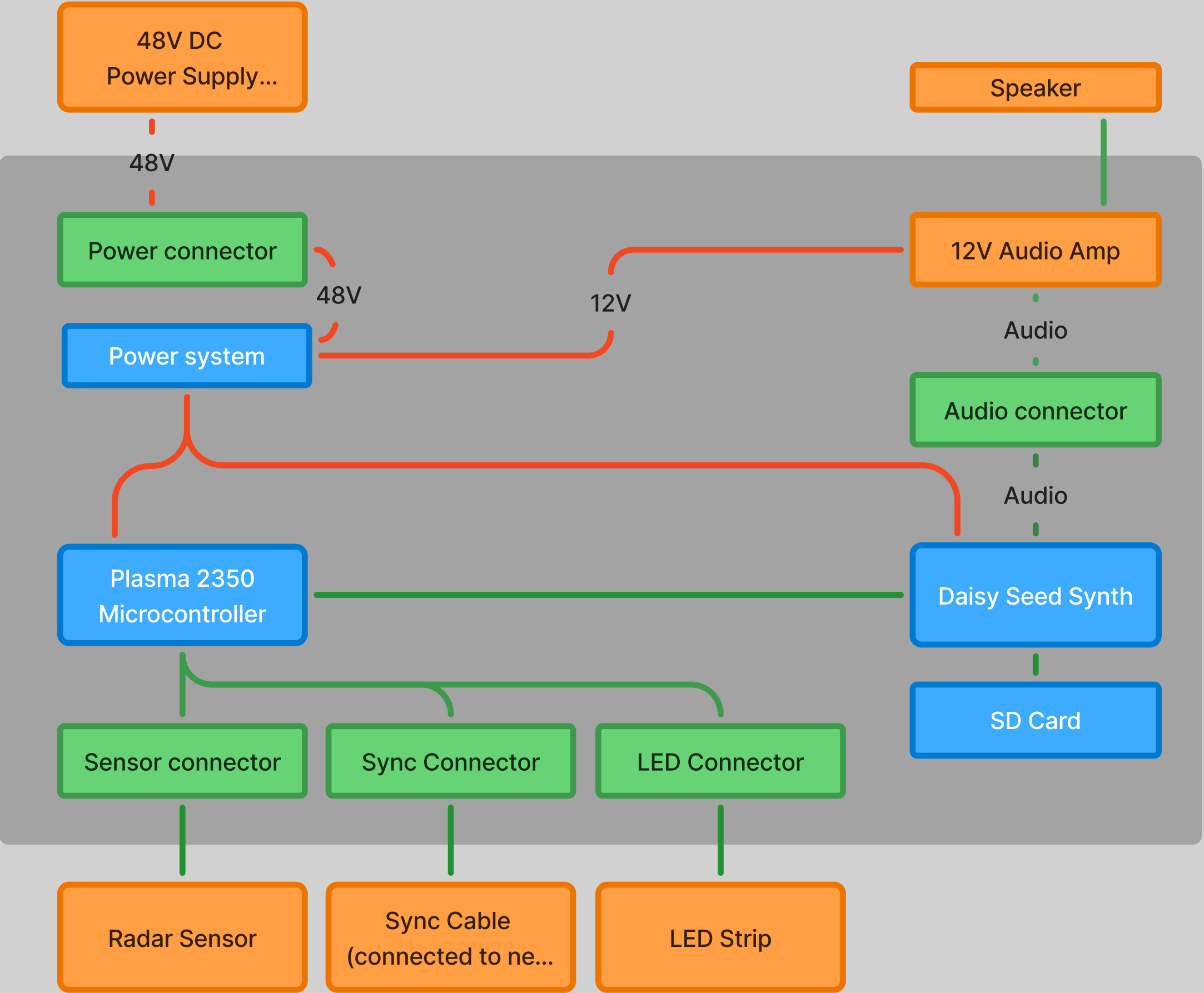


2m
LED Strip



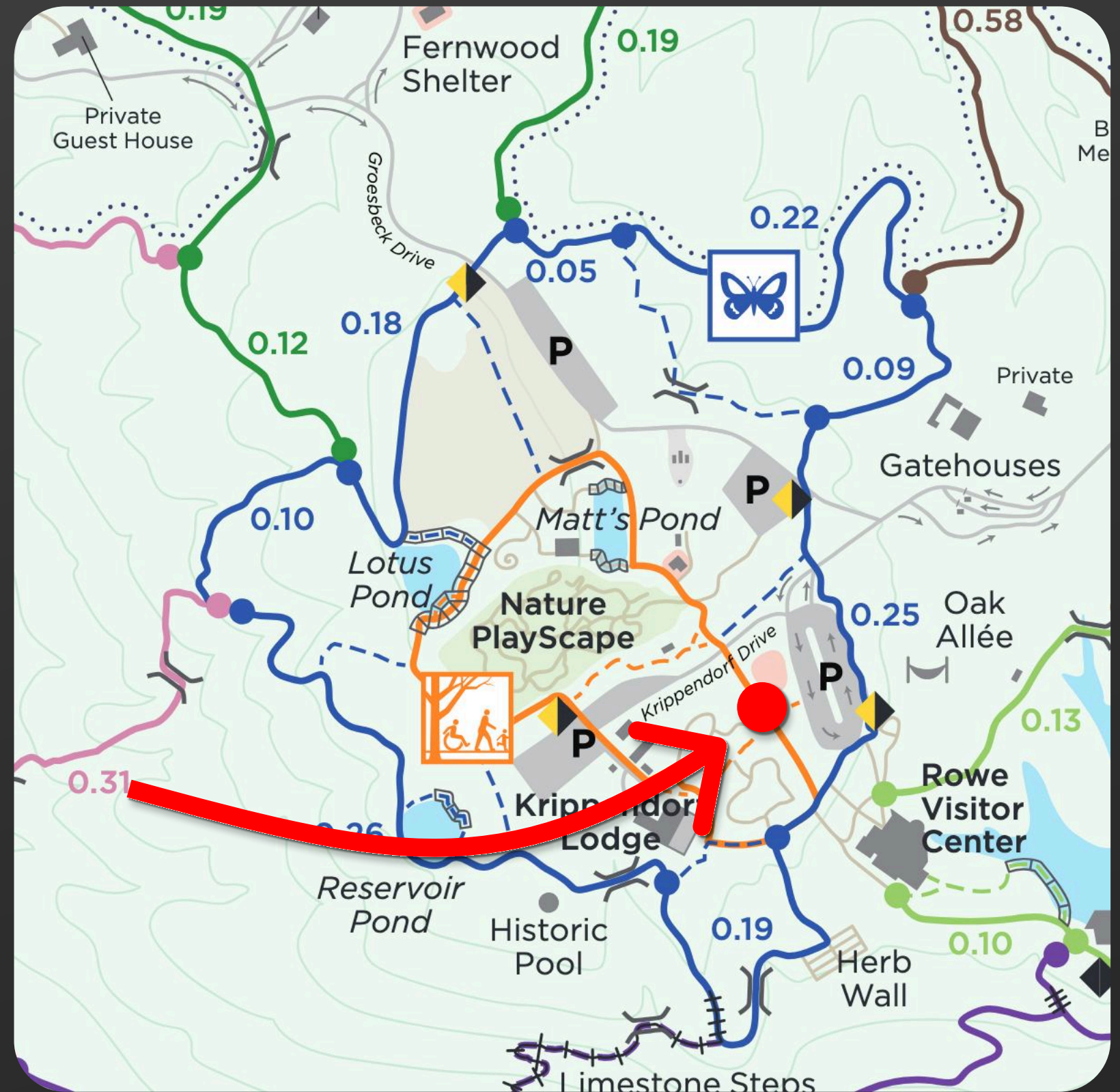
Forms may vary
significantly depending
on the specific materials sourced.

EXAMPLE CIRCUIT BOARD LAYOUT



APPROXIMATE LOCATION

Near the visitors center
parking lot.





Herm Greider

Retired Software Developer with 40 years experience leading, designing, and coding all types of software systems for defense and commercial applications. BS Computer Science, UC. MS CS Wright State. Currently prototyping lighting and sound projects.

<https://www.linkedin.com/in/hermgreider>



Aaron Greider

Graphic Designer/Front-End Web Developer at Jungle Jim's International Market. BS Communication Design from UC DAAP 2022. Cincinnati Nature Center Member. Eagle Scout and native plant enthusiast. Bane of honeysuckle.

BUDGET

Total with 8 new Portals,
paper maché motifs,
featured improvements,
and insurance: **\$4172**

*This number is negotiable
depending on the final number
of additional portals*

Category	Item Name	Per Item	Number needed	Total
Electronics Upgrade				
	Custom PCBs (16+spares)	100	1	100
	Audio jacks	1	16	16
	SD card holder	4	16	64
	Large electronics box	8	16	128
Sync Feature (requires Electronics Upgrade)				
	RS-485 modules for sync	2	16	32
	RS-485 connectors	1	16	16
	Cat-5 cable (200 ft)	60	1	60
	Level converters	2	16	32
Power Upgrade				
	Mean Well 48V HLG-320H-48 IP67 320W power supplies	50	2	100
	48V to 12V 10A converter	15	16	240
	12V to 5V converter	10	16	160
	Power cable	50	1	50
Audio Upgrade (requires Power Upgrade)				
	Waterproof connectors for power snake	40	1	40
	Hiletgo TPA3118 audio amps	7	16	112
	Outdoor-rated small speakers	30	16	480
New Units - electronics (8) (old units will reuse 2025 parts)	Connectors and cable	50	1	50
	Plasma	20	8	160
	Daisy Seed	30	8	240
	Adafruit Neopixel 2m LED strips	30	8	240
	Micro SD	5	8	40
	Small electronics box (sensor)	1	8	8
	Sensors	4	8	32
	Headers, connectors, cables, solder	60	1	60
New Units - portal assembly (8)				
	Windows	10	8	80
	Acrylic window panes	35	8	280
	Acrylic cover for frame	40	8	320
	Frame	50	1	50
	Band clamps	2	16	32
	Fence posts	5	16	80
	Caulk, paint, screws	100	1	100
	Netting	20	1	20
Misc				
	Insurance	500	1	500
	Transportation	250	1	250
Total				
				4172

PROJECT PLAN

June	Early Prototype — 12" x 12" frosted pane, one LED strip with proximity and synth
June 30, 2026	Acquire windows and electronics
July 31, 2026	Finalize appearance
August 31, 2026	Construction of Portal frames
September 31, 2026	Portal software programming
October 31, 2026	Finishing touches and Backyard Halloween Demo
November 13, 2026	Installation
January, 2027	Teardown

PORTFOLIO

light.greider.org/aaron

light.greider.org/herm

Follow our progress on [Instagram](#)

REFERENCES

Lynsey Ficker *Graphic Designer at Cincinnati Nature Center*

+1 (513) 325-8566

fickerlynsey@gmail.com

Linwood Cobb *Creative Director at Jungle Jim's*

+1 (513) 340-2417

zcobb@junglejims.com

Keith Reinig *Scoutmaster, Scout Troop 530*

+1 (937) 681-4362

keith.reinig@gmail.com